

PROTECTION PLAN

Future Mall Capstone - Task 3

Protecting the mall's most exposed systems & its people

Field	Value
Project	Future Mall - Shopping for Tomorrow (Capstone)
Document	Task 3 - Protection Plan
Scope	Table for 3 mall systems: Free Wi-Fi, Cashier Device, Mall Website
	Awareness sign with 3 simple tips for employees
Audience	Mall staff, IT team, and management
Format	PDF (this file)

Prepared by the Future Mall team - Capstone Project 2026.

Protection Table - Three Mall Systems

The three most exposed systems of Future Mall are the public Free Wi-Fi network, the cashier devices that handle daily sales, and the mall website customers use to browse and order. For each system the table lists its main risk and the protection method applied to reduce that risk.

Mall System	Main Risk	Protection Method
Free Wi-Fi	Rogue devices / 'evil twin' access points trick users into connecting to a fake network and stealing their passwords and payment details (man-in-the-middle).	Use WPA2/WPA3 enterprise authentication with per-user login; encrypt all traffic with captive-portal HTTPS plus a VPN option for sensitive browsing; isolate the guest network with VLAN segmentation so guests cannot reach mall devices; block peer-to-peer client traffic (client isolation).
Cashier Device	POS malware or a USB key infect the cashier, and staff leaving the device logged in lets others modify prices or leak customer payment data.	Lock the cashier to a single restricted POS user, auto-lock after 90 seconds; whitelist USB devices and scan them on insertion; keep the OS and POS software updated, run up-to-date antivirus, and disable side-loaded payments; record an audit log of every transaction and price change.
Mall Website	SQL injection / XSS let attackers deface the site, steal customer logins, or inject phishing pages; brute force on the admin panel.	Use HTTPS with TLS across the whole site; parameterise every SQL query; validate and encode all user input; enforce strong admin passwords (12+ chars) with MFA and IP allow-listing; keep CMS and plugins updated; take nightly backups and do weekly restore tests.

The table above is the deliverable table for Task 3. Every system has at least one realistic risk and one concrete protection method.

Awareness Sign - 3 Simple Tips for Employees

A poster/sign design that will be hung in the mall's staff areas (break room and back office). It contains three simple, easy-to-remember security tips.

Awareness Sign

FUTURE MALL - STAY SAFE, STAY SMART

1

Think before you click

Never open links or attachments from unknown senders - if it is urgent, verify on another channel.

2

Lock, then walk

Lock your screen whenever you leave your device - even for a minute. An unattended cashier is an open door.

3

Never share your password

No real IT team will ever ask for your password or OTP.
Keep codes private, use a password manager and MFA.

Read the full three-system protection table on the previous page. For anything unusual, contact the IT help desk immediately.